

CAREER: Foundations of Self-Adaptive GPU Graph Mining Across Algorithms, Datasets, and Multi-GPU Architectures

Overview

Graph mining (finding cliques, motifs, dense subgraphs, and structural patterns) underpins scientific discovery, cybersecurity, fraud detection, computational biology, and the data preparation pipelines of modern AI systems. Yet graph mining remains the least scalable form of graph computation: it combines irregular memory access, combinatorial search, deep pruning, and unpredictable output sizes. GPUs offer the parallelism and bandwidth this workload demands, but today’s GPU graph-mining systems are hand-tuned for one algorithm, one dataset, and one GPU generation. Three structural problems result: (1) a *reproducibility crisis*, where reported speedups rarely transfer across datasets or hardware, making fair comparison nearly impossible; (2) a *performance-portability gap*, where code optimized for V100 collapses on A100/H100/B200 because launch configurations, shared-memory layouts, and scheduling assumptions all shift across architectures; and (3) a *single-GPU ceiling*, where scaling a mining algorithm to multi-GPU or out-of-memory graphs typically requires a near-complete rewrite.

This CAREER project develops the algorithmic and systems foundations of a self-adaptive, GPU-native graph-mining stack: a small set of composable primitives (neighbor intersection, candidate expansion, pruning predicates, pattern verification, output-sensitive enumeration, and task-based load balancing), an architecture-aware auto-tuning layer that learns from hardware and graph features, and a runtime that scales a single program from one GPU to many without user rewriting. The intellectual stance is that graph-mining *primitives*, not whole frameworks, are the right unit of reuse: as AI-assisted programming systems increasingly automate code generation, the bottleneck shifts from writing CUDA to providing reliable, composable, performance-portable abstractions that future coding agents can compose into correct and efficient mining systems: the bricks and mortar from which higher-level mining systems are built.

The PI will pursue three integrated, layered research thrusts:

- T1. GPU-native primitives and a unified programming abstraction.** Identify the small set of recurring building blocks behind diverse mining algorithms (triangle/clique enumeration, k -core/ k -truss, motif and subgraph matching, quasi-clique mining, butterfly counting, and temporal-motif mining) and express them as composable, GPU-friendly primitives with well-defined warp-, block-, and device-level mappings. A declarative interface lets developers describe what to mine while the framework decides how.
- T2. Architecture-aware auto-tuning for performance portability.** Cast tuning as a structured search problem parameterized by hardware features (SM count, L2/shared-memory size, HBM bandwidth, warp size, tensor-core capability) and graph features (degree skew, density, locality). Use lightweight learning-based models to predict near-optimal launch configurations, tile sizes, and frontier thresholds, refined by in-situ profiling. The goal: a single mining program runs near hand-tuned performance across V100, A100, H100, and Blackwell-class GPUs without user intervention.
- T3. A unified single-GPU to multi-GPU adaptive runtime.** Treat mining as task-parallel search-tree exploration, not graph partitioning. Design NVLink/NVSwitch-aware work stealing, host-memory and NVMe spill strategies for graphs that exceed aggregate GPU memory, and an automatic single-GPU-to-multi-GPU scaling path so users do not rewrite code to scale out. Together, T1–T3 establish the missing systems layer that makes GPU graph mining usable, reproducible, and portable, and that gives the next generation of AI-assisted developers a principled

vocabulary for building graph-mining systems at scale.

Intellectual Merit

The project advances graph-mining systems and high-performance computing in three ways. **First**, it formulates and validates a *primitive-level abstraction* that separates the logic of mining (expand, prune, verify, emit) from the mechanics of GPU execution (memory layout, warp scheduling, kernel choice), demonstrating that a small, well-chosen set of GPU-native primitives suffices to express modern mining algorithms efficiently. **Second**, it formulates performance portability as a learnable, structured search problem over a joint hardware–graph feature space and develops auto-tuning algorithms that exploit mining-specific structure (search-tree imbalance, neighborhood access patterns) which generic tuners miss. **Third**, it casts multi-GPU mining as a task-parallel search-tree problem with work stealing, communication-aware scheduling, and memory coordination, providing the first runtime in which a single mining program scales seamlessly from one GPU to many. Expected outcomes include open-source software, a community benchmark suite, and new algorithmic and systems results published in top venues (SIGMOD, VLDB, SC, PPOPP).

Broader Impacts

The project democratizes high-performance graph analytics by removing the expert-tuning barrier that confines GPU graph mining to a small community of CUDA specialists, and prepares the field for an era in which AI-assisted programming agents compose mining systems from trustworthy primitives. The integrated education and outreach plan includes: (i) a new upper-level/graduate course on *GPU Systems for Data-Intensive Computing* at Rowan University, using the framework as a teaching platform so students can do GPU mining research without months of CUDA prerequisites; (ii) curriculum modules introducing parallel graph algorithms into existing undergraduate database and algorithms courses; (iii) a reproducible community benchmark of mining algorithms $\times$ datasets $\times$ GPU architectures, pursued in collaboration with the ACM SIGMOD reproducibility initiative; (iv) mentoring of graduate and undergraduate researchers at Rowan, with explicit recruitment of women and students from groups historically underrepresented in computing through Rowan’s CS Diversity initiatives and CRA-WP/REU programs; (v) an annual summer workshop introducing high-school students from the South Jersey public school system to data science and GPU computing; and (vi) industry engagement with NVIDIA and AMD graph teams to position the framework as an evaluation platform for new hardware. All software, benchmarks, teaching materials, and lecture recordings will be openly released to maximize community uptake.

Keywords: graph mining; GPU computing; programming abstractions; performance portability; auto-tuning; multi-GPU runtime systems; reproducibility.